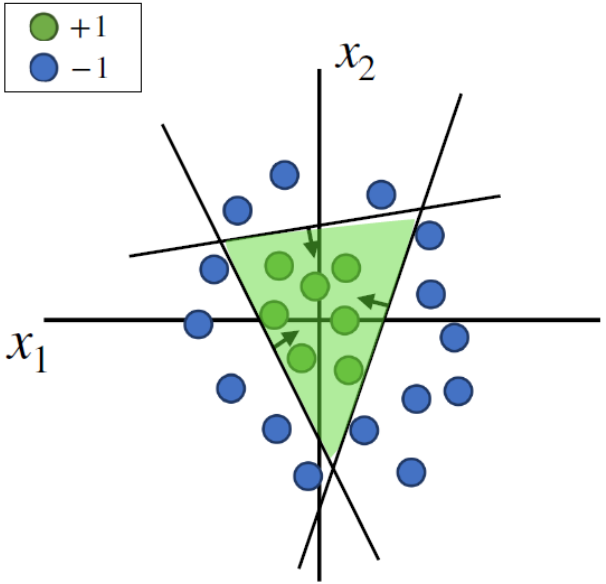
Session 7: Dimensionality reduction

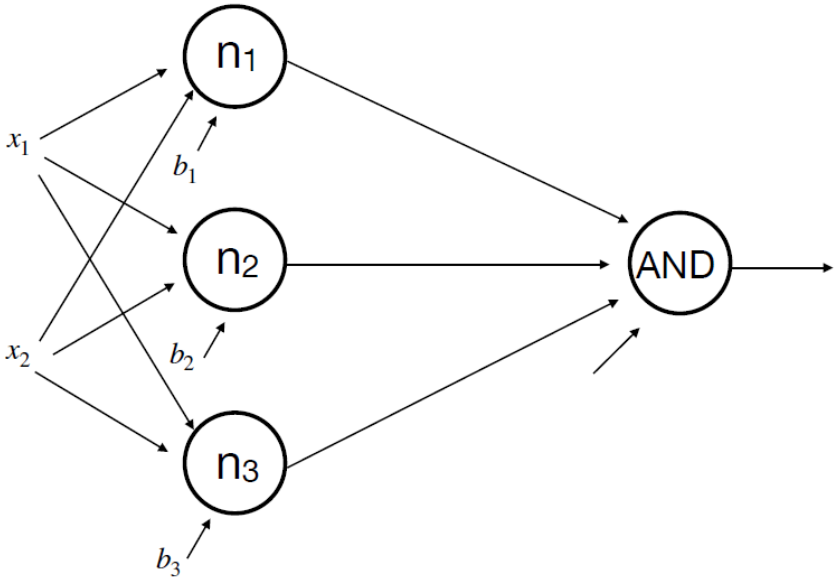
BSCAI 2026

Javier Muro

Recap

Geometric view



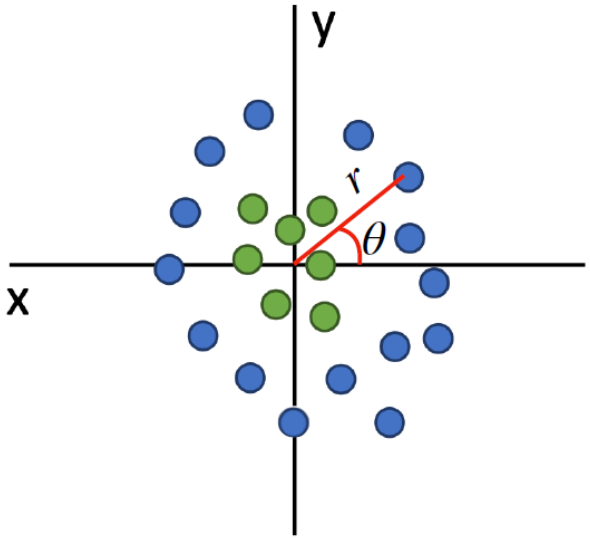


decision boundary of the AND neuron

2 layers of neurons!

How else can we solve this?

Original space

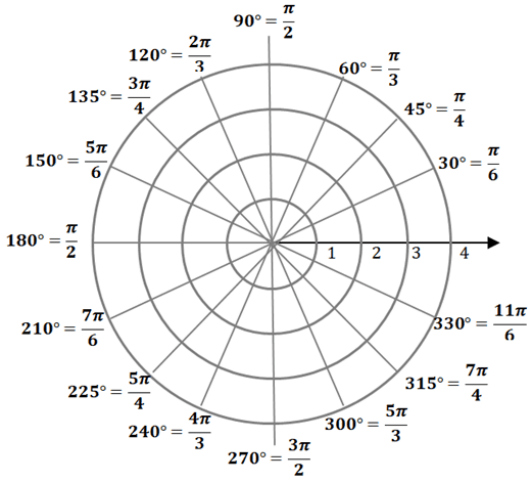


polar coordinates 🐼

$$r = (x^2 + y^2)^{1/2}$$

$$\theta = \tan^{-1}(y/x)$$

Feature transform

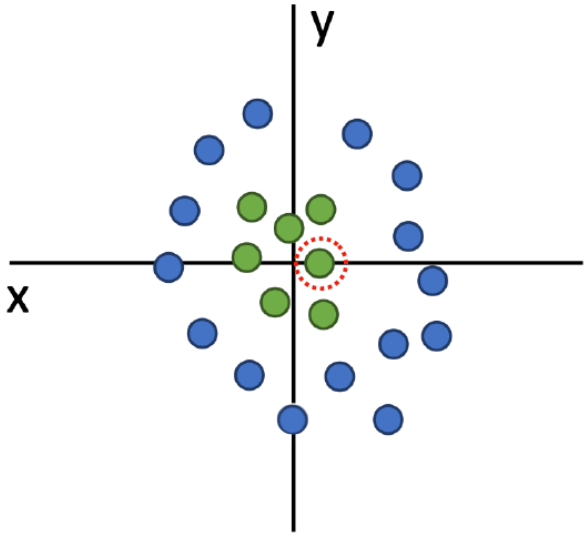


?

Adapted from: J. Johnson (UMich)

One solution: **Feature Transforms**

Original space



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$$r = (x^2 + y^2)^{1/2}$$

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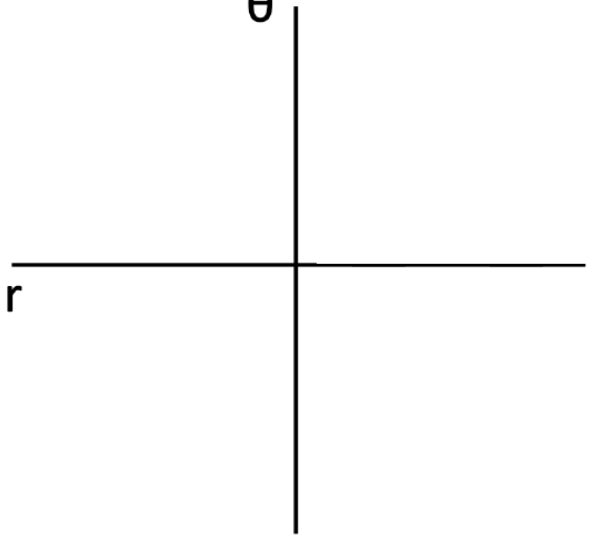
Feature transform

Feature space

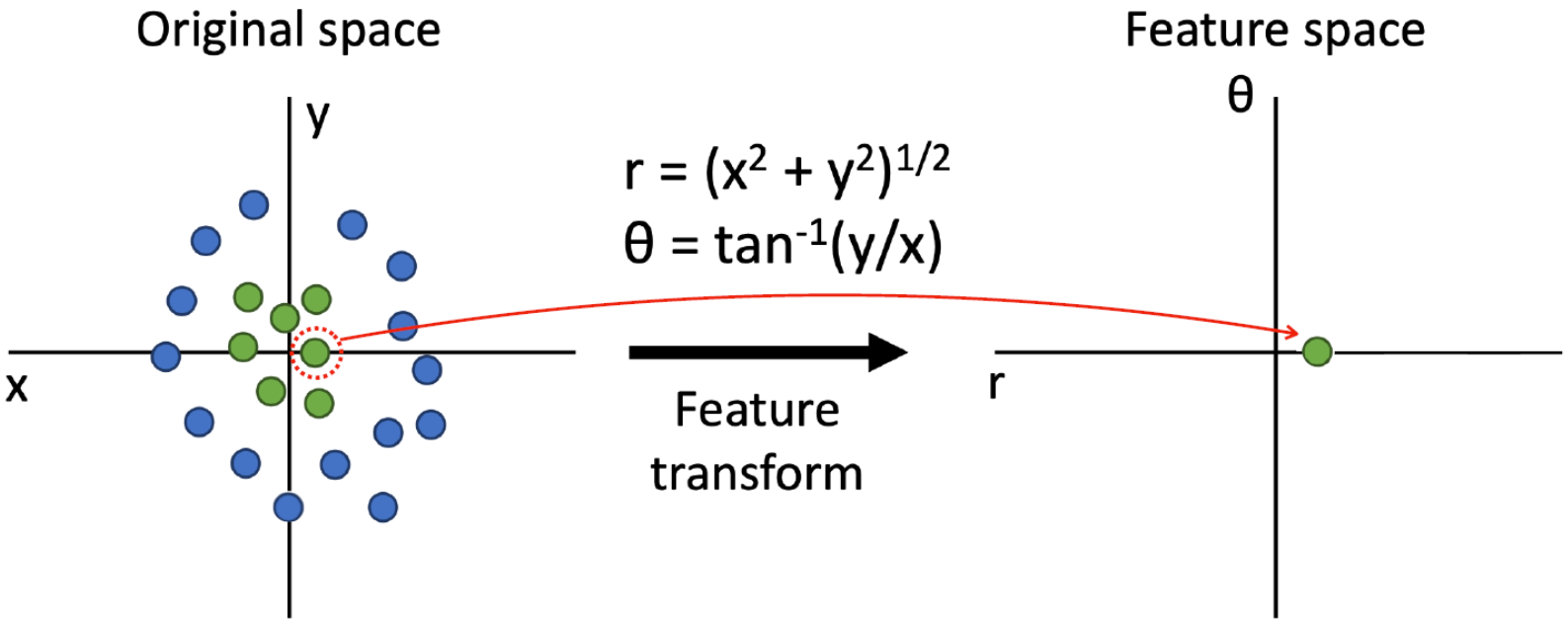


takes in a point in (x,y) → outputs a point in (r,θ)

e.g. (0.25,0) → ???

Adapted from: J. Johnson (UMich)

One solution: **Feature Transforms**

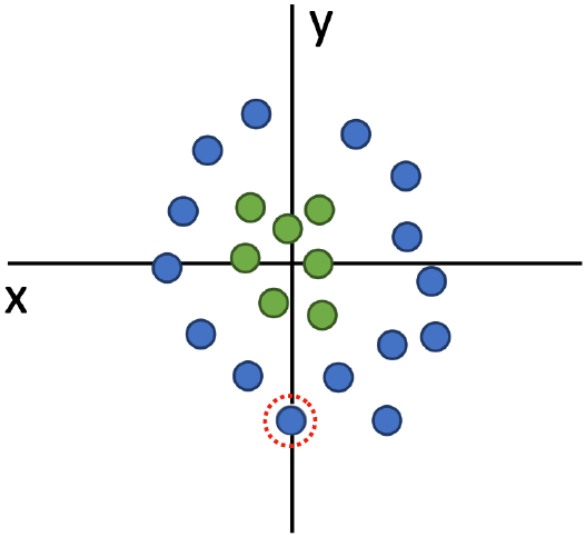


takes in a point in (x,y) → outputs a point in (r,θ)

e.g. (0.25,0) → (0.25,0)

One solution: **Feature Transforms**

Original space

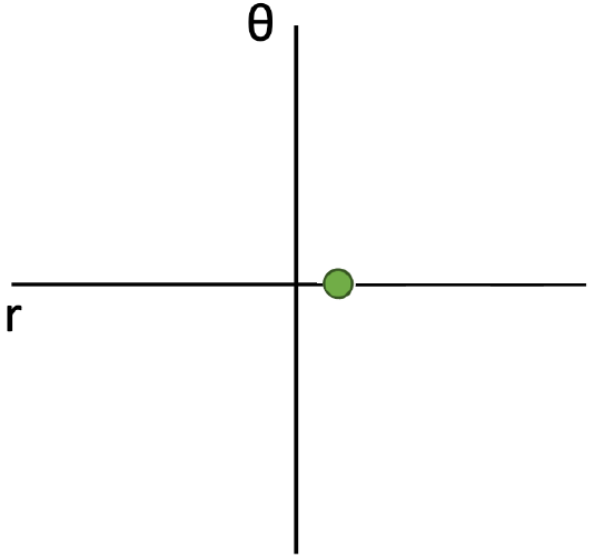


$r = (x^2 + y^2)^{1/2}$

$\theta = \tan^{-1}(y/x)$

Feature transform

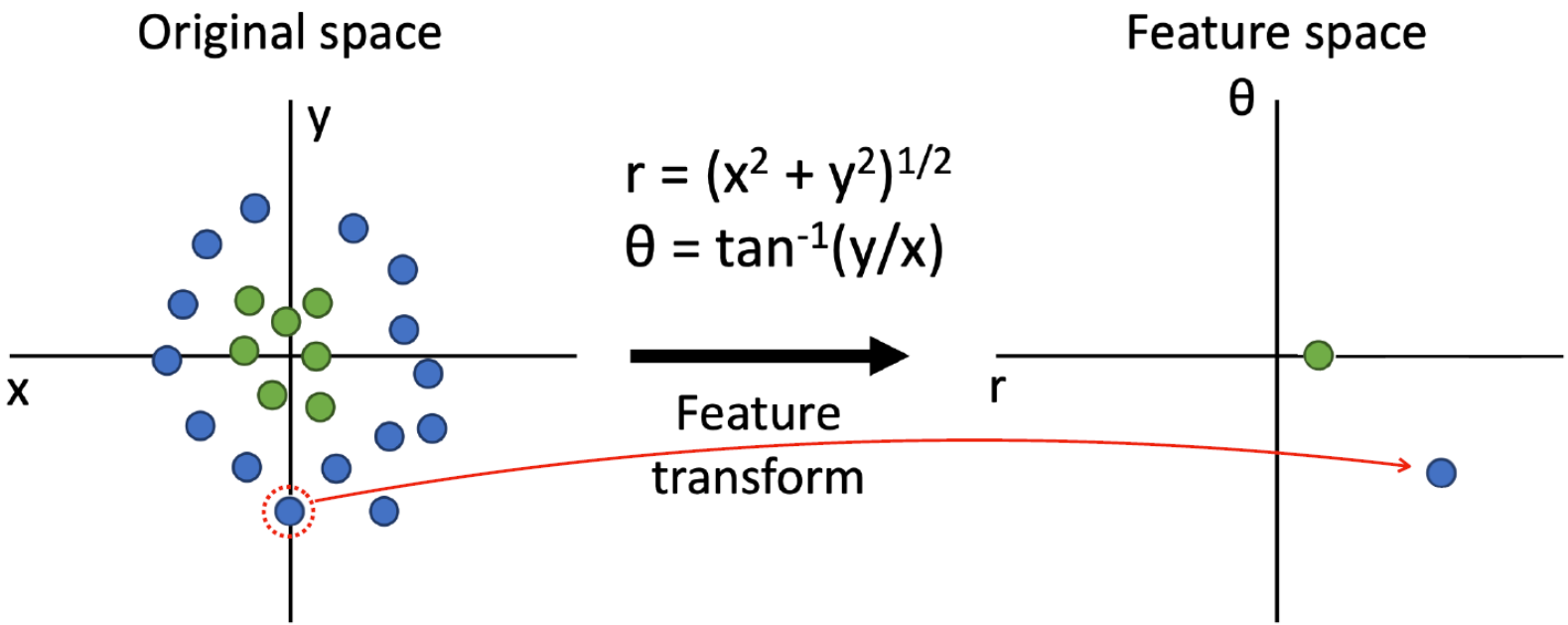
Feature space



takes in a point in (x,y) → outputs a point in (r,θ)

e.g. $(0, -1)$ → ???

One solution: **Feature Transforms**

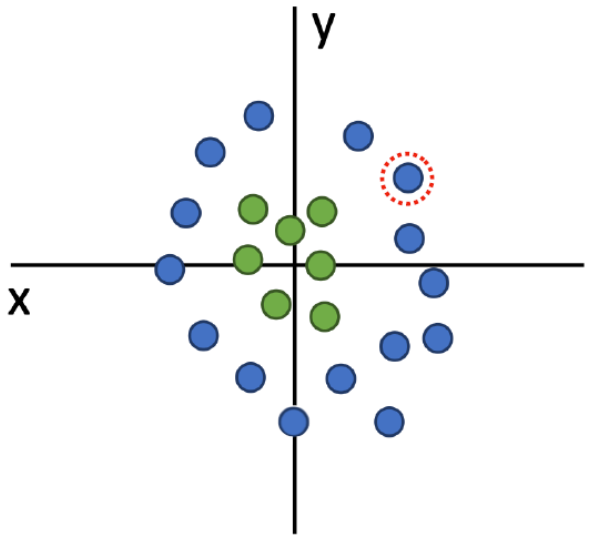


takes in a point in (x,y) → outputs a point in (r,θ)

e.g. $(0, -1) \rightarrow (1, -\pi/2)$

One solution: **Feature Transforms**

Original space



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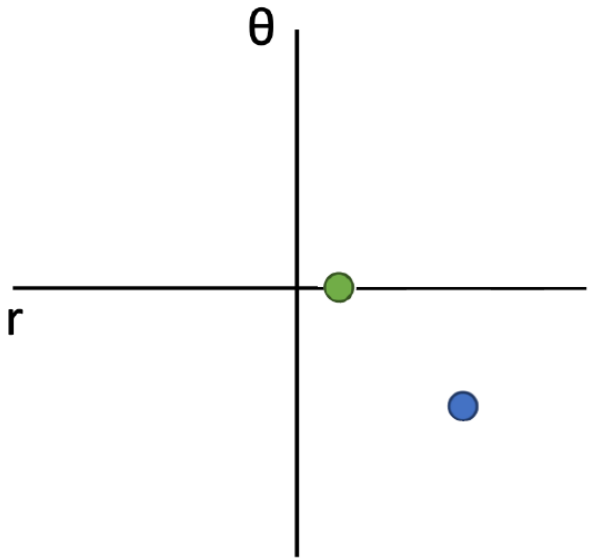
Feature transform

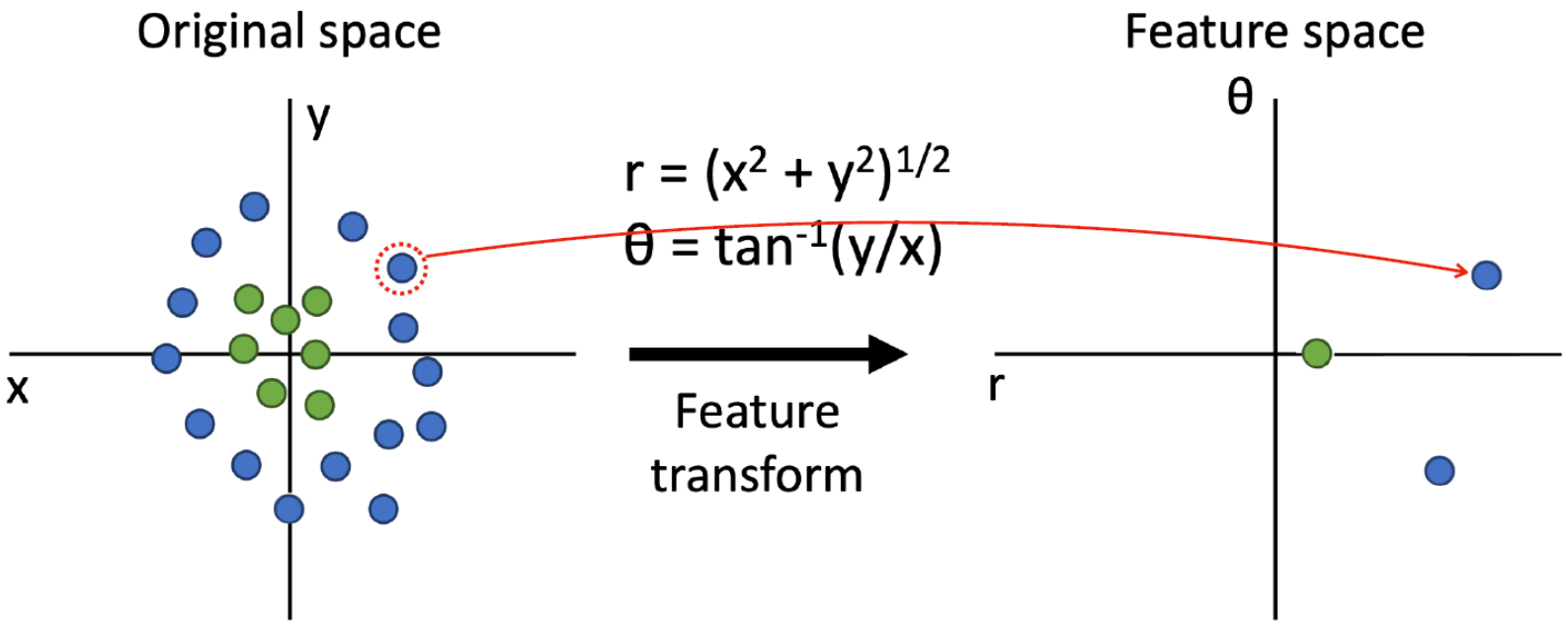
Feature space



takes in a point in (x,y) → outputs a point in (r,θ)

e.g. (1,1) → ???

One solution: **Feature Transforms**



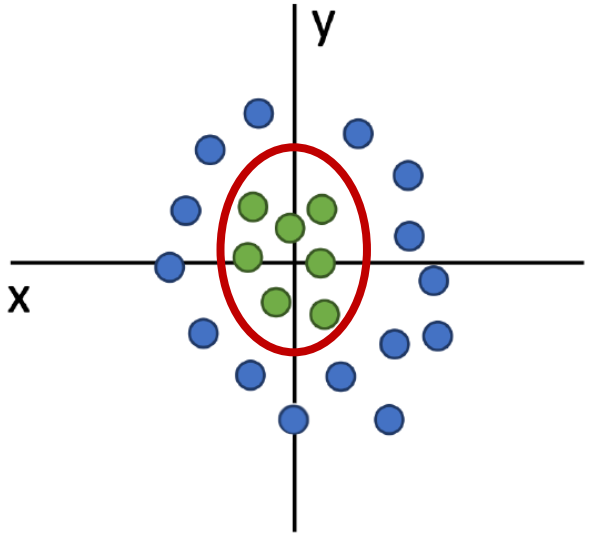
takes in a point in (x,y) → outputs a point in (r,θ)

e.g. $(1,1)$ → $(\sqrt{2}, \pi/4)$

A linear classifier applied to transform data is like a non-linear classifier applied to raw data

One solution: **Feature Transforms**

Original space

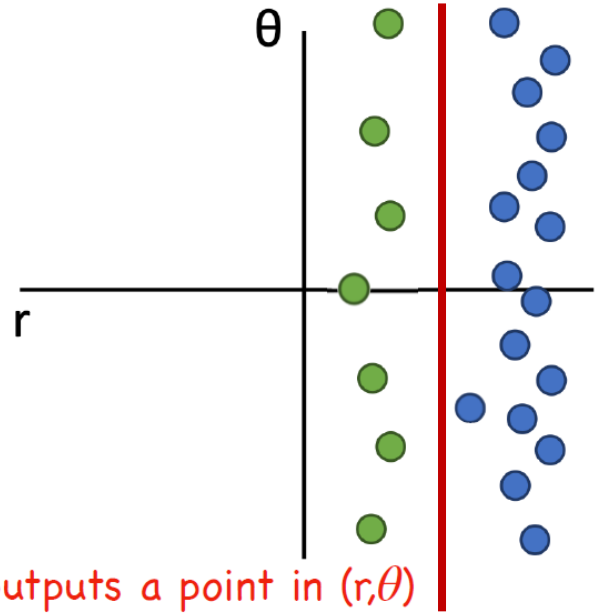


$r = (x^2 + y^2)^{1/2}$

$\theta = \tan^{-1}(y/x)$

Feature transform

Feature space



takes in a point in (x,y)

outputs a point in (r,θ)

etc…

Adapted from: J. Johnson (UMich)

Question:

- What other feature-space transformations do you know?
- AKA latent-space transformations

Do we need a quick recap on PCA and classical dimensionality reduction?

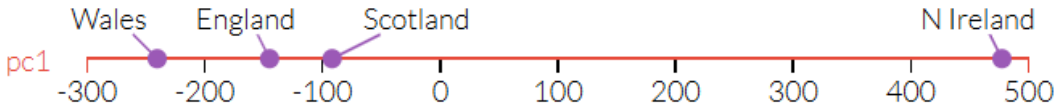
PCA recap

https://setosa.io/ev/principal-component-analysis/

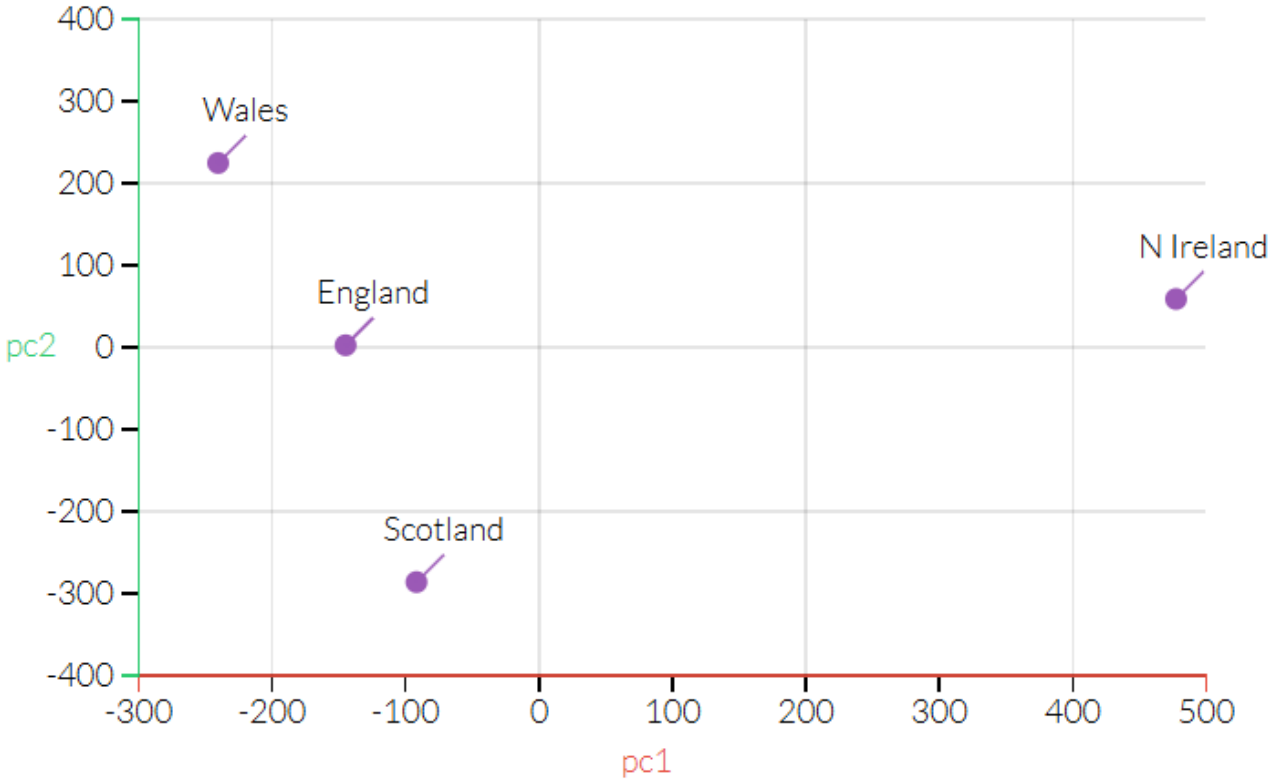
We want to find vectors of maximum variance

Average consumption of 17 types of food in grams per person per week for every country in the UK

	England	N Ireland	Scotland	Wales
Alcoholic drinks	375	135	458	475
Beverages	57	47	53	73
Carcase meat	245	267	242	227
Cereals	1472	1494	1462	1582
Cheese	105	66	103	103
Confectionery	54	41	62	64
Fats and oils	193	209	184	235
Fish	147	93	122	160
Fresh fruit	1102	674	957	1137
Fresh potatoes	720	1033	566	874
Fresh Veg	253	143	171	265
Other meat	685	586	750	803
Other Veg	488	355	418	570
Processed potatoes	198	187	220	203
Processed Veg	360	334	337	365
Soft drinks	1374	1506	1572	1256
Sugars	156	139	147	175

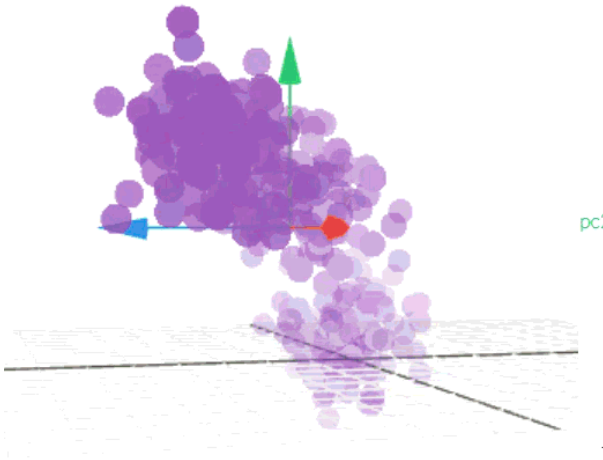


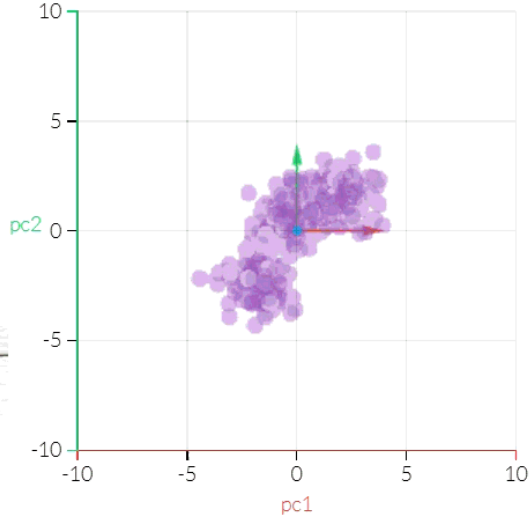
PCs are ortogonal to the previous one

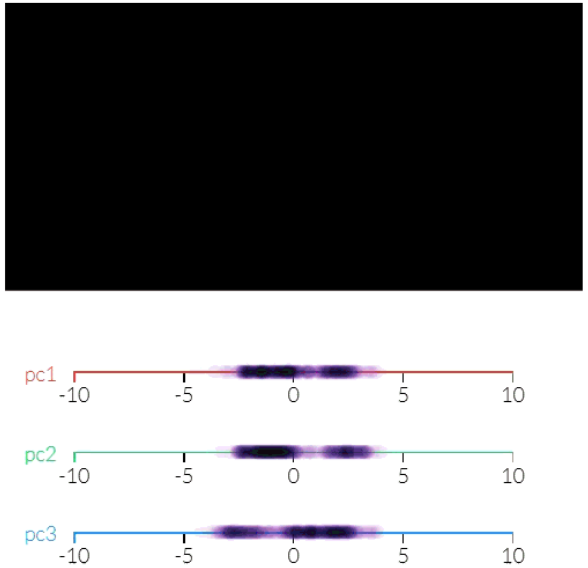


Find vectors of maximum variance

https://setosa.io/ev/principal-component-analysis/







show PCA reset

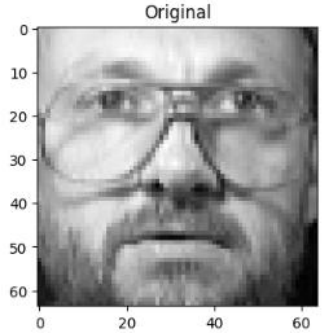
To get PC scores, we must project the observations onto the eigenvectors.

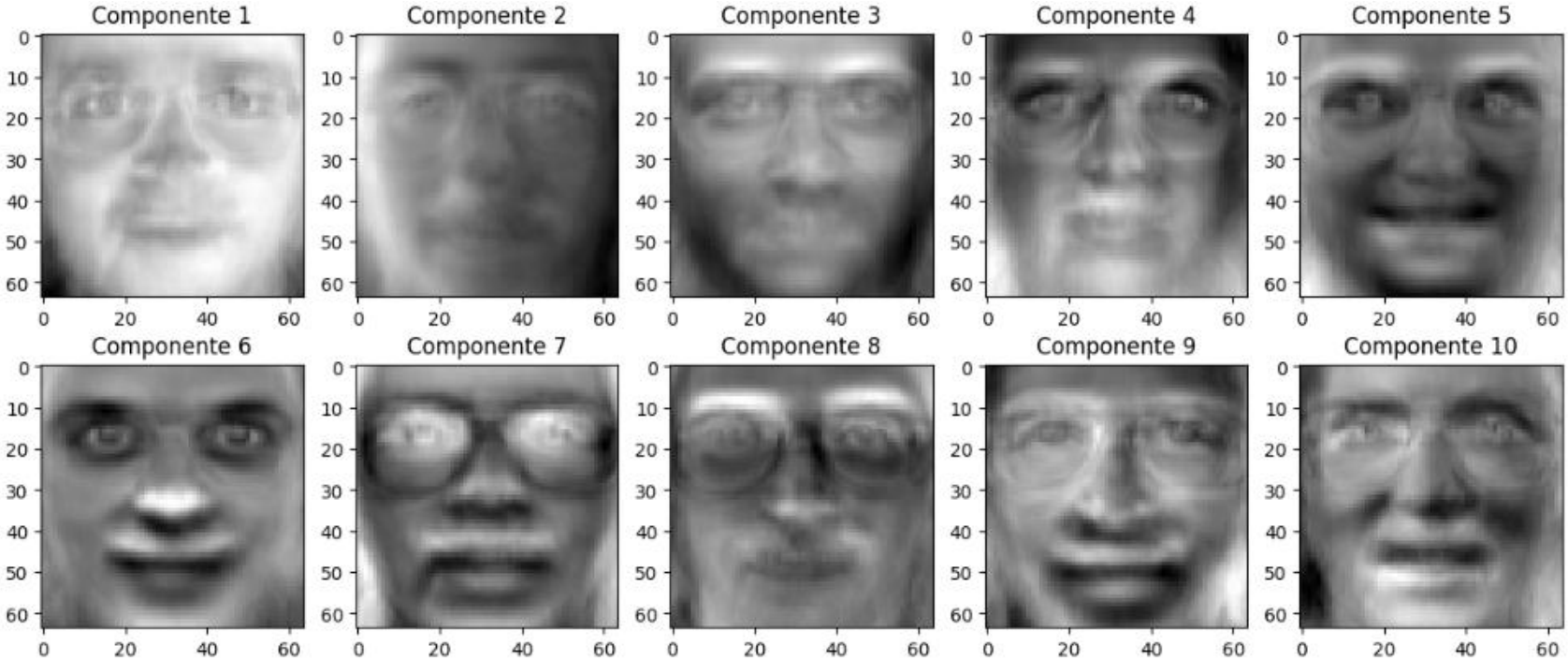
Eigenvectors: direction variance
Eigenvalues: how much variance is along the eigen vector

https://setosa.io/ev/eigenvectors-and-eigenvalues/

PCA in images

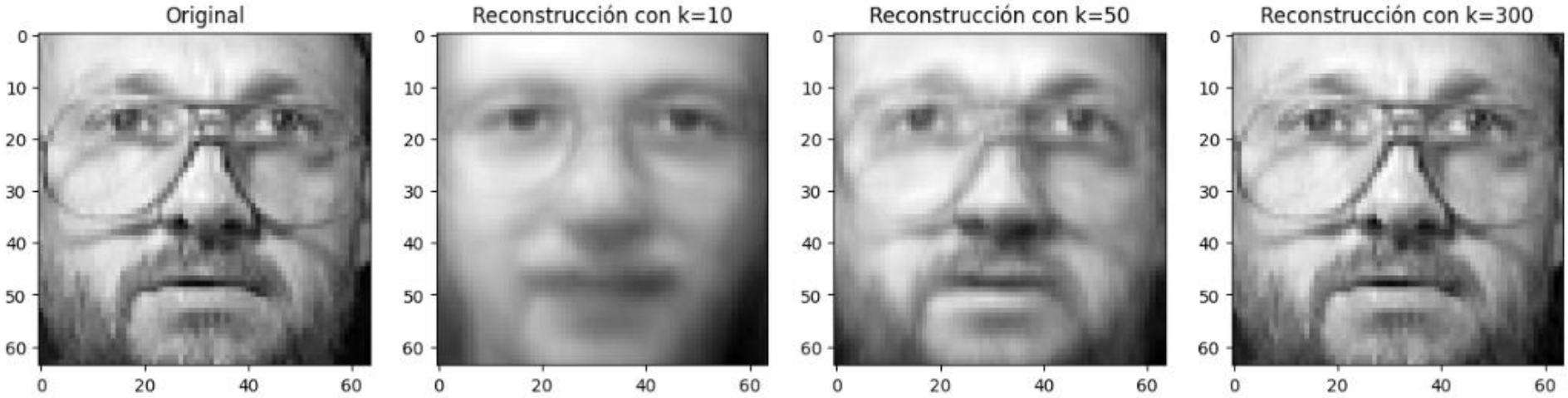
- from sklearn.datasets import fetch_olivetti_faces



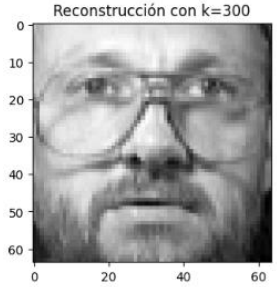


PC in images

- And can be reconstructed



Hidden layers perform similar operations

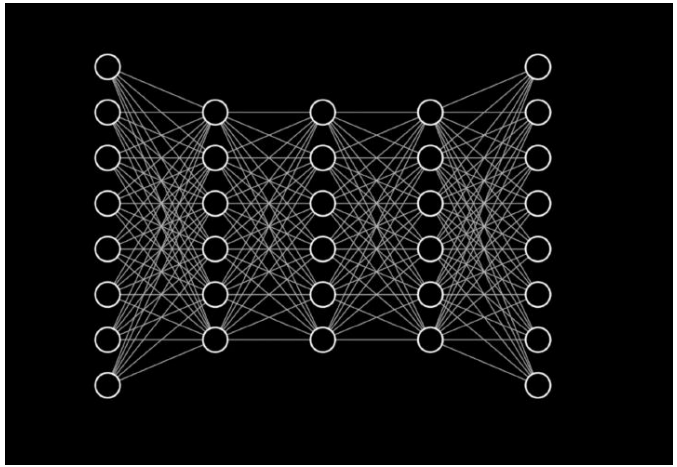


with some differences:

- PCA is unsupervised, linear separation and maximizes variance
- NN transformations are supervised and maximize class separation

- But both learn data structures and reduce their dimensionality (encoding)
- And transformed data can be reconstructed (decoding)

Bottom-up from pixels to edges, then to object parts, and finally to object identity, the network effectively performs dimensionality reduction by compressing high-dimensional raw data into low-dimensional abstract concepts



In facial recognition, the machine can "discover" the latent highlights or features essential for identifying a face without being explicitly told what they are

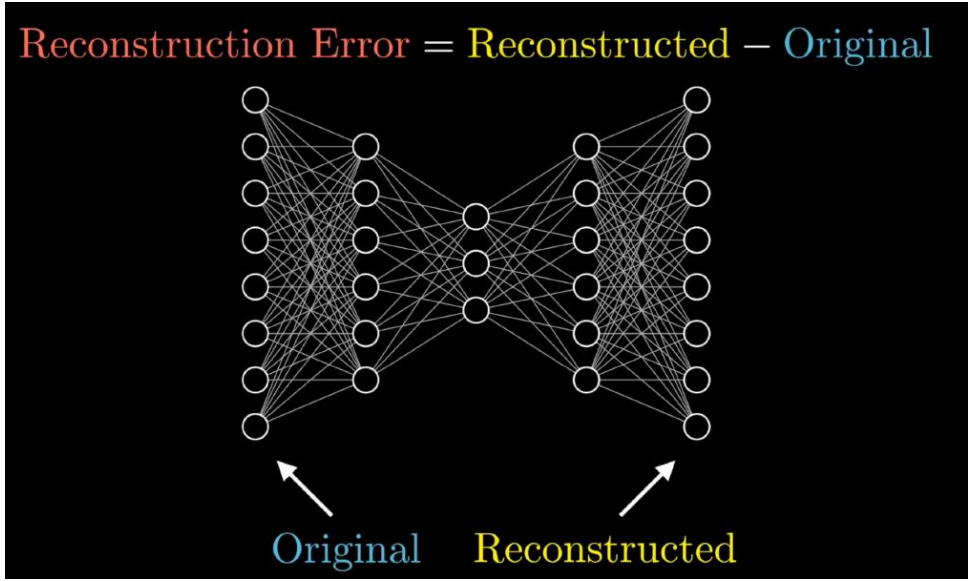
Autoencoders

NN **hidden layers**: compress to simplify classification

Encoders: compress to build a compact representation, task agnostic

Hidden layers recode into only useful discriminative features

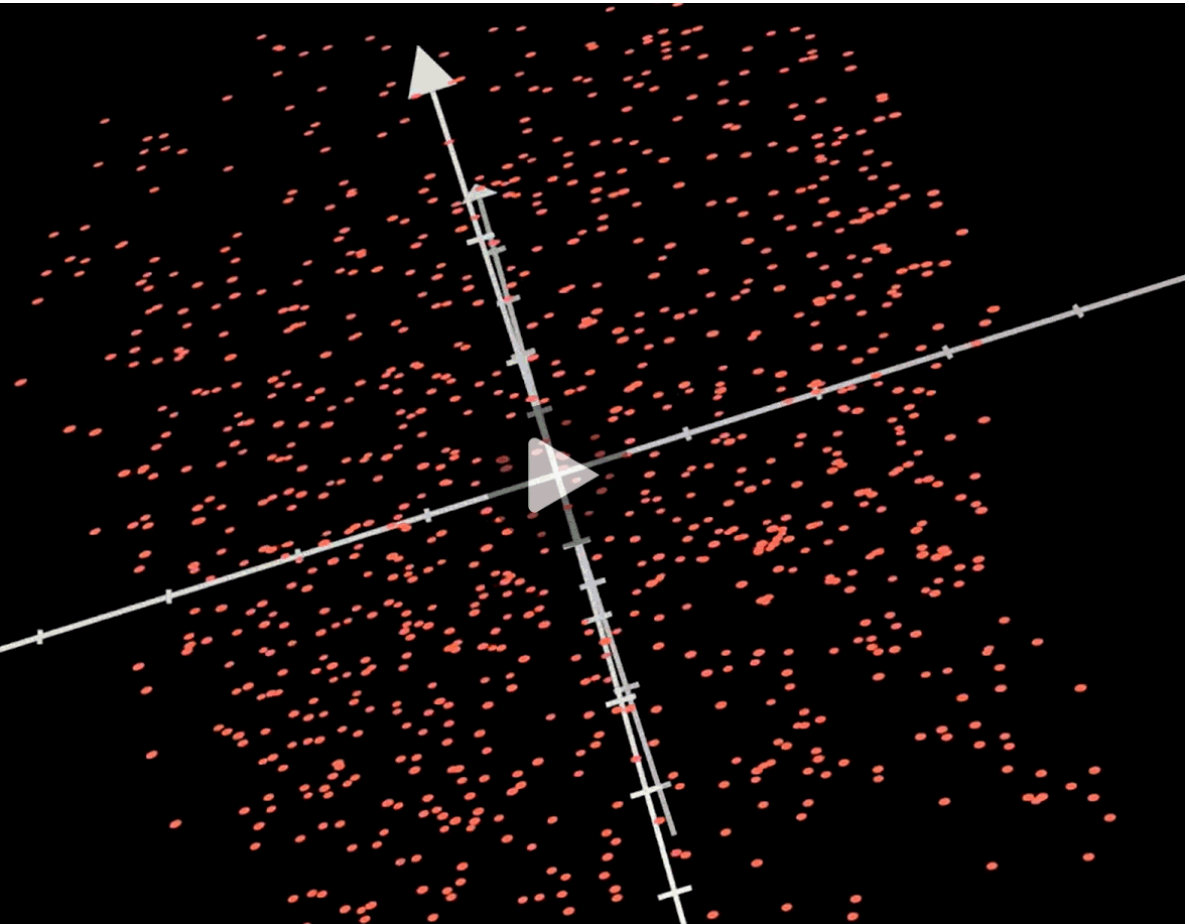
Decoders recode back everything into data space



Not for prediction, but to learn the latent feature space/structure, everything that matters for understanding the input, not just predicting specific labels (and then maybe improve predictions with it).

Simple Explanation of AutoEncoders

Latent space without structure

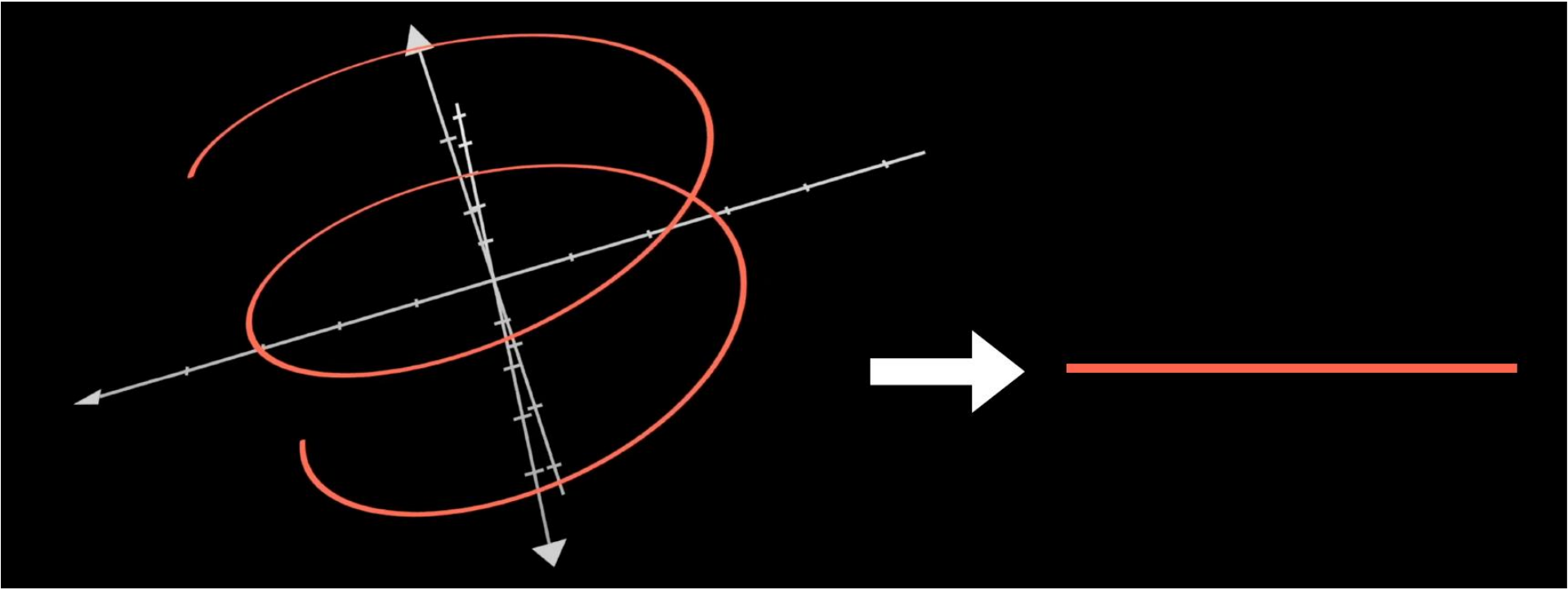


Points are here distrubuted randomly across space. We would need three predictors/axes/coordinates to represent each of them.

But data in nature has structure. NN learn that structure to constrain the information and find solutions efficiently.

Simple Explanation of AutoEncoders

Constraiend data in the same 3D space



We don't need 3 coordinates if we transform the data. Only the position along the spiral.

Latent space demo

- https://wikipedia2vec.github.io/demo/

Feature space in the brain

- **Feature Extraction**: Much like Deep Learning, the brain extracts low-level features (pixels/edges) and combines them into increasingly abstract "latent" representations (objects/identities).
- **Dimensionality Reduction:** In both biological systems and **autoencoders**, high-dimensional sensory data is compressed into a low-dimensional "bottleneck" (latent representation) that retains only the essential features needed to reconstruct or recognize the stimulus
- **Multimodal Integration:** The **hippocampus** acts as an integrator, weaving disparate sensory fragments—the sight of a cup, the smell of coffee—into a unified episodic memory
- **Prediction and Inference:** The brain uses **predictive coding**, generating internal expectations about future inputs and using the "error signal" to update synaptic weights.

Perception = Prediction + Error Correction

Feature space in the brain

- **Hebbian Learning**: The fundamental rule for resculpting feature spaces is "neurons that fire together, wire together," strengthening synaptic links through Long-Term Potentiation (LTP).
- **Sparse Coding**: The brain utilizes sparse distributed coding, where a very small, highly selective set of neurons (concept cells) responds to specific complex stimuli like a celebrity's face, instead of firing all the neurons involved in face recognition.

How do humans generalize?

- Similarity based judgement: humans assess how closely a new situation resembles a **familiar** one in a psychological scape relying on propotypes

- Rule based application learnt through **induction**

- **Abstract structure mapping**: we identify the underlying abstract relational structure, rather than just its concrete features

How do humans generalize?

Two stage process:

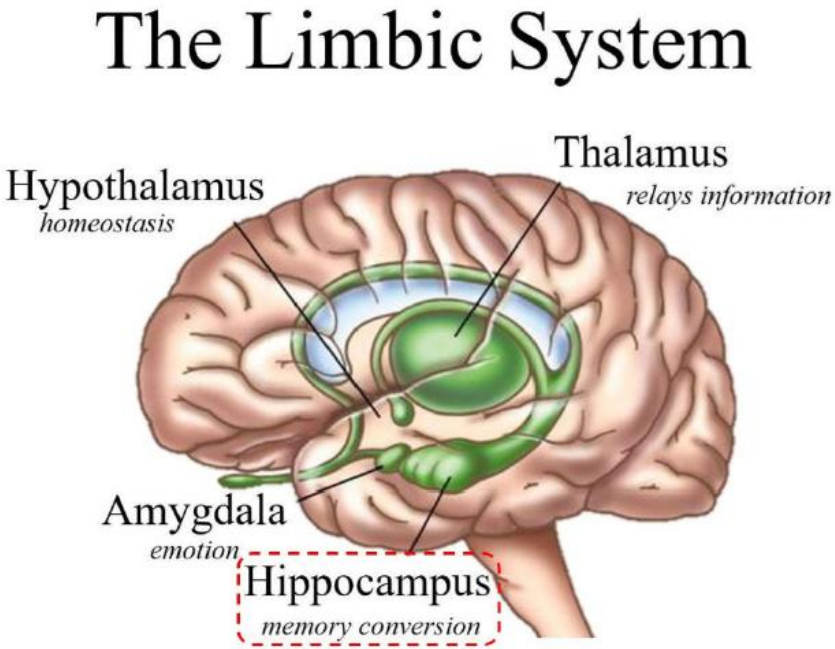
1.- Determining **relevance**: the brain automatically determines what features are relevant.

2.- Refining **representations**: Through experience, mental representations of these features become more precise, allowing for more efficient, abstract and accurate future generalizations.

Opposite of Deep learning: the better we learn something, the better we can abstract and generalize to novel scenarios

Neural components

Hippocampus: rapid, episodic generalization is thought to rely here, which reactivates and integrates existing memories with new experiences



Cognitive components

Metageneralization: we can flexibly switch between different strategies such as **compositional** (breaking down a problem, useful in *pseudocode*) or **holistic** (conjunctive, e.g. fase recognition) approaches depending on the structure of the task.

Human-like systematic generalization through a meta-learning neural network

How do humans generalize?

Brenden M. Lake & Marco Baroni

Nature **623**, 115–121 (2023) | Cite this article

https://www.nature.com/articles/s41586-023-06668-3

Cognitive components

Efficient coding: we aim for mental efficiency by distilling information into simple, essential forms, often priorizing a small subset of features (color, shape of ripe fruit) to guide decisions.

Humans learn generalizable representations through efficient coding

Zeming Fang & Chris R. Sims

Nature Communications **16**, Article number: 3989 (2025) | Cite this article

https://www.nature.com/articles/s41467-025-58848-6

Recalling past experiences

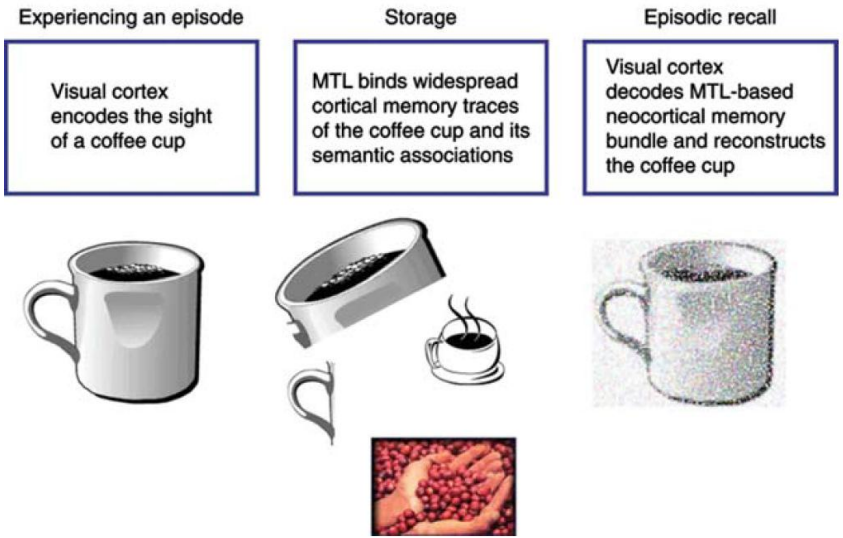


FIGURE 7.4 Episodic learning of conscious events. The sight of a specific coffee cup is encoded in visual maps for different features, like location, color, line orientation, motion, and so on. In the medial temporal lobe (MTL) the separate feature maps may be "bound" together into a single "gestalt." Notice that the coffee cup is associated with a handful of raw coffee beans and with the smell of hot coffee. During slow wave sleep, the active memory of the coffee cup is recoded into lasting synaptic changes in the cortex. Recall from episodic memory reverses these steps. *Source: Baars, with permission.*

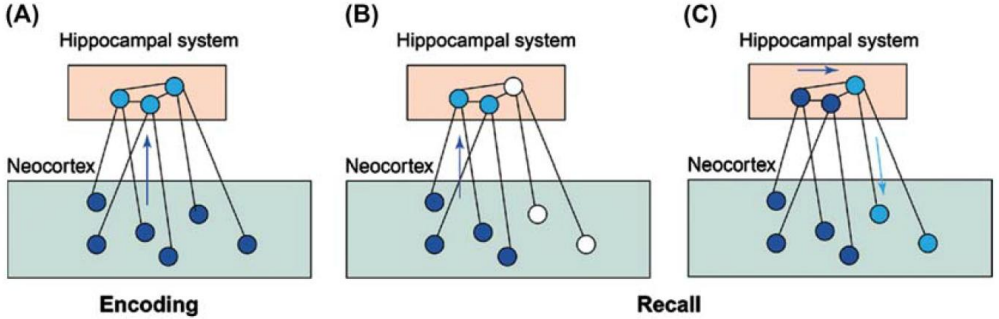


FIGURE 7.5 Hippocampus and cortex. The hippocampal area supports temporary episodic memories that activate and are stored in the cortex. In this cartoon, we show stages of encoding and recall and the roles of the hippocampal system and the cortex. (A) An episodic memory is formed (Encoded) in the cortex and gives rise to a pattern of activity with the hippocampal system. (B) During Recall, part of the pattern is active in the cortex, reactivating part or all of the hippocampal pattern. (C) The reactivation of the hippocampal pattern will, in turn, reactivate the remaining cortical pattern. *Source: Gluck et al. (2003).*

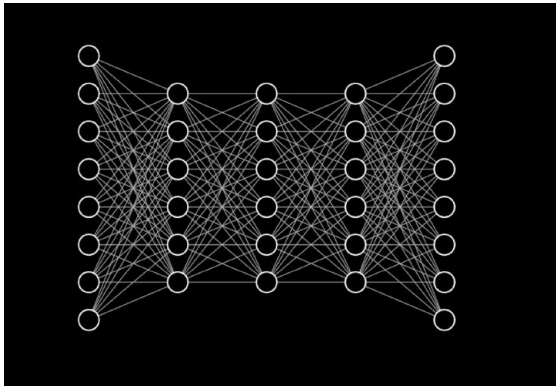
Fundamentals of Cognitive Neuroscience. Nicole Gage and Bernard Baars.

Invariance: A core challenge for CNN is selectivity/invariance, where the system must identify an object (like a face) despite changes in angle, lighting, or position.
Our brain learns patterns that might be efficient in some circumstances, but deceivable in others.

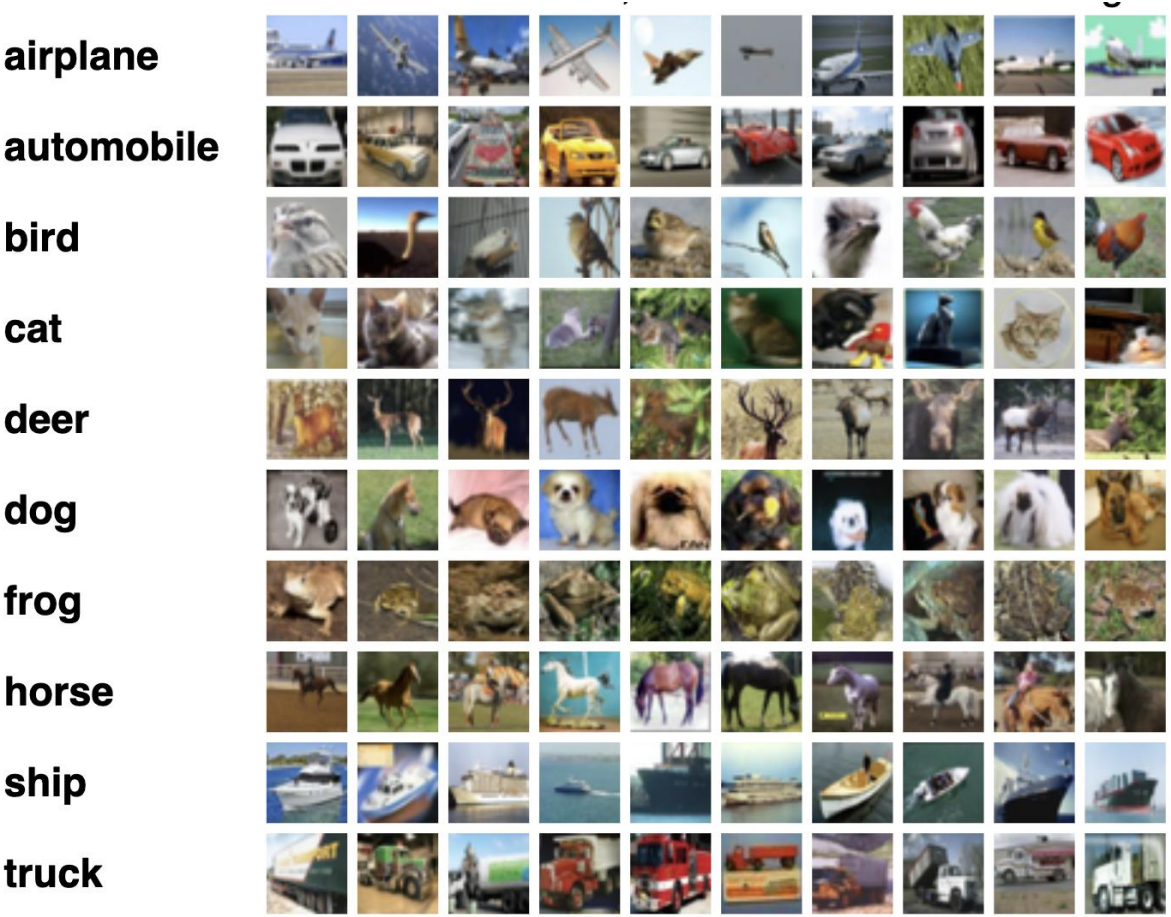
Python time!

1 Train a CNN

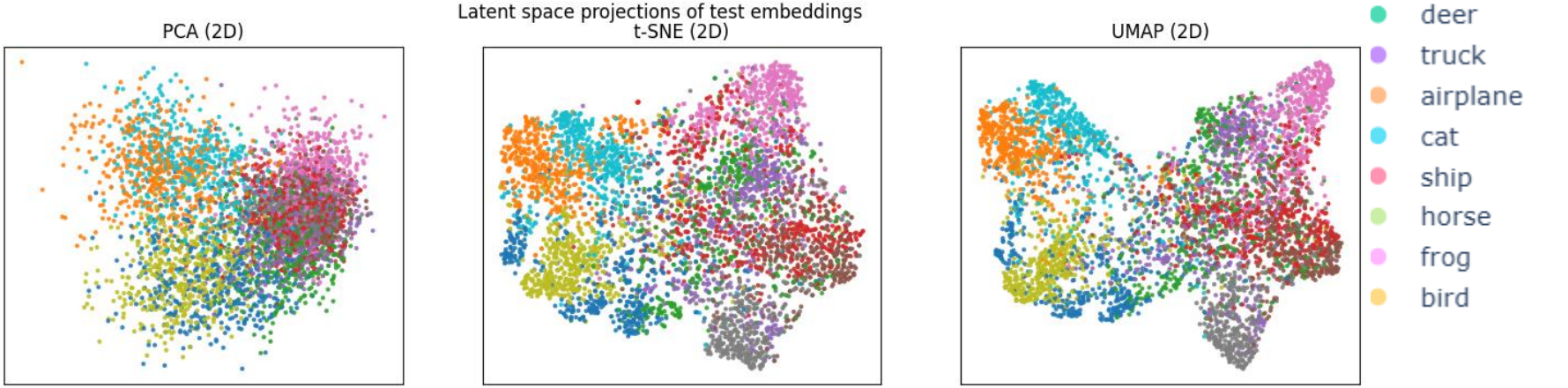
2 Open the last layer and get the embedding



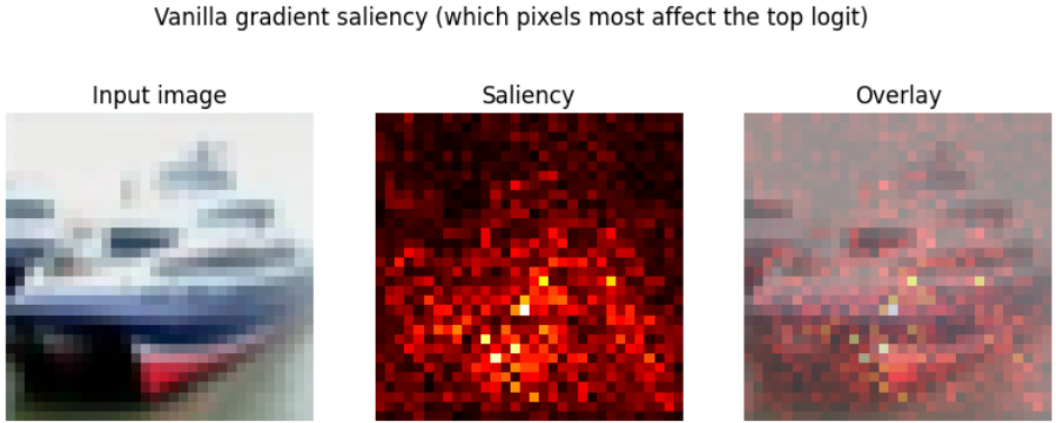
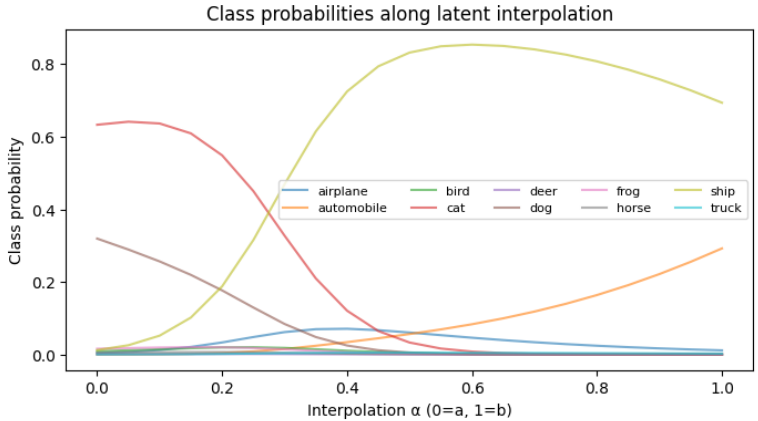
CIFAR-10



3 Visualize embeddings using:



4 Linear probes

- Run a classification on the embedding, vs on the raw input.
  - Which one is faster?
  - which one is more accurate?

5 See how the model thinks and mapped the latent space.